Jay Niteshbhai Vekariya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.38
Ashadeep Vidhyalay, Surat.	Class XII (Gujarat Secondary and Higher Secondary Education Board, Gandhinagar)	2019	79.23%
P. P. SAVANI CHAITANYA VIDYA SANKUL, Mota Varachha Abrahma Road, Surat.	Class X (Gujarat Secondary and Higher Secondary Education Board, Gandhinagar)	2017	79.17%

INDUSTRY INTERNSHIP EXPERIENCE

Micron Technology - MCP Server & AI Agent Engineering Intern

- Built a conversational analytics platform (**Angular, FastAPI, PostgreSQL, Claude LLM via Model Context Protocol**) letting product managers query application metrics in natural language.
- Engineered an **MCP-driven Text-to-SQL agent** converting questions into SQL with **prompt engineering, memory, and guardrails**.
- Developed a **TypeScript SDK** and **REST APIs** to ingest user-journey events from multiple apps into **PostgreSQL**.

PROJECTS

RouterDC on Indic Benchmarks - Query-Based LLM Routing GitHub

- Evaluated **RouterDC** query-based routing across **8 frozen LLMs** from 2B to 32B on **30,482 Hindi and low-resource Indic MCQs**, training only an 86M mDeBERTa-v3 encoder against precomputed scores with **dual contrastive losses** over top-3/bottom-3 models and 5 k-means question clusters.
- Across **5 seeded runs and 2 ablations**, routing **matched the best single model** (39.59% vs. 39.61%), indicating that the evaluated candidate pool limited routing gains.
- **Tech Stack:** Python, PyTorch, Transformers, mDeBERTa-v3, contrastive learning, scikit-learn, NumPy

Agentic Enterprise Intelligence - AI Incident Investigator GitHub

- Built an AI incident-investigation agent in Python that answers root-cause questions over an enterprise corpus and flags any claim it cannot ground in a source, **hybrid retrieval fusing BM25 with dense embeddings by reciprocal rank, then a cross-encoder rerank, lifted Recall@5 from 0.800 to 0.940**.
- Engineered a **LangGraph state machine** over an **MCP server of 17 tools** spanning GitHub, logs, and SQL, where **the single write tool halts behind a graph interrupt for human approval,** retrieved text is neutralized against 7 prompt-injection classes, with 302 tests including planted adversarial documents.
- **Tech Stack:** Python, LangGraph, MCP, FastAPI, SSE, Ollama, sentence-transformers, BM25, SQLite, Docker, pytest

Movie Recommendation System GitHub

- Built a **content-based movie recommender** on the TMDB 5000 dataset, combining genres, keywords, top-5 billed cast, director, and plot overview into a tag corpus of **4,806** movie records.
- Generated **7,000-dimensional bag-of-words** features using **CountVectorizer** and ranked movies using **cosine similarity**, with a **Streamlit** app returning the top 10 recommendations and posters via the **TMDB REST API**.
- **Tech Stack:** Python, Pandas, Scikit-learn, Streamlit, TMDB API

SKILLS

Programming Languages: C, C++, Python

Databases: MySQL, PostgreSQL

Web Technologies: HTML, CSS, JavaScript, TypeScript

Frameworks & Libraries: FastAPI, Angular

Tools & Platforms: Git, Docker, Wireshark

AI Agents & LLM Technologies: Model Context Protocol (MCP), LangChain, LangGraph, RAG, n8n, Hugging Face

Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, Transformers, RNN, LSTM

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Soft Skills: Problem Solving, Effective Communication, Teamwork

SCHOLASTIC ACHIEVEMENTS

Secured **5th Place** in the IIT-H Deep Learning 2026 Hackathon on Kaggle

Secured **3rd Rank** in the Information Technology Department, B.Tech. 7th Semester

Solved **475+** Data Structures and Algorithms problems across LeetCode and GeeksforGeeks

RELEVANT COURSES

Deep Learning	Visual Computing	Advanced Computer Networks	Foundations of Machine Learning
Fraud Analytics	Operating Systems	Object-Oriented Programming	Computer Organization & Architecture
Network Security	Artificial Intelligence	Database Management Systems	Advanced Algorithms and Data Structures

EXTRACURRICULAR

Founder, AgentifyAI - Startup Studio, IIT Hyderabad

- Selected among the **Top 24 startups** for IIT Hyderabad's 60-day Startup Studio program, prototyped an AI agent marketplace for SMBs.